Compositional monitoring of energy-mix drift on the simplex

Which carrier did the structural work?

Standard compositional data analysis — what happens when we read the simplex (shares that sum to one) forward in time.

Follow along on the repository — the slide deck, manuscript, and live projector are all open.

H^s runs any compositional dataset the CoDa community can describe — every view here is reproducible on your data.

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github.com / PeterHiggins19 / higgins-decomposition

Community folder: CODA-Association/ · Hand-out in UN-6 locales (EN · FR · ES · RU · ZH · AR)

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

The size view hides the work

A carrier can be small in share yet do most of the structural work — its share of the squared compositional motion.

Germany electricity, 25 years
the size view — the standard share / stacked-area chart

Coal and lignite recede. Nuclear phases out.
Gas holds. Solar and wind grow steadily.
Wind passes coal in the late 2010s.

What the size view misses

Germany Solar, 2005 → 2006

```
starting share           0.21 %
structural Power Share  71.1 %
Activation Coefficient   ≈ 333 ×
```

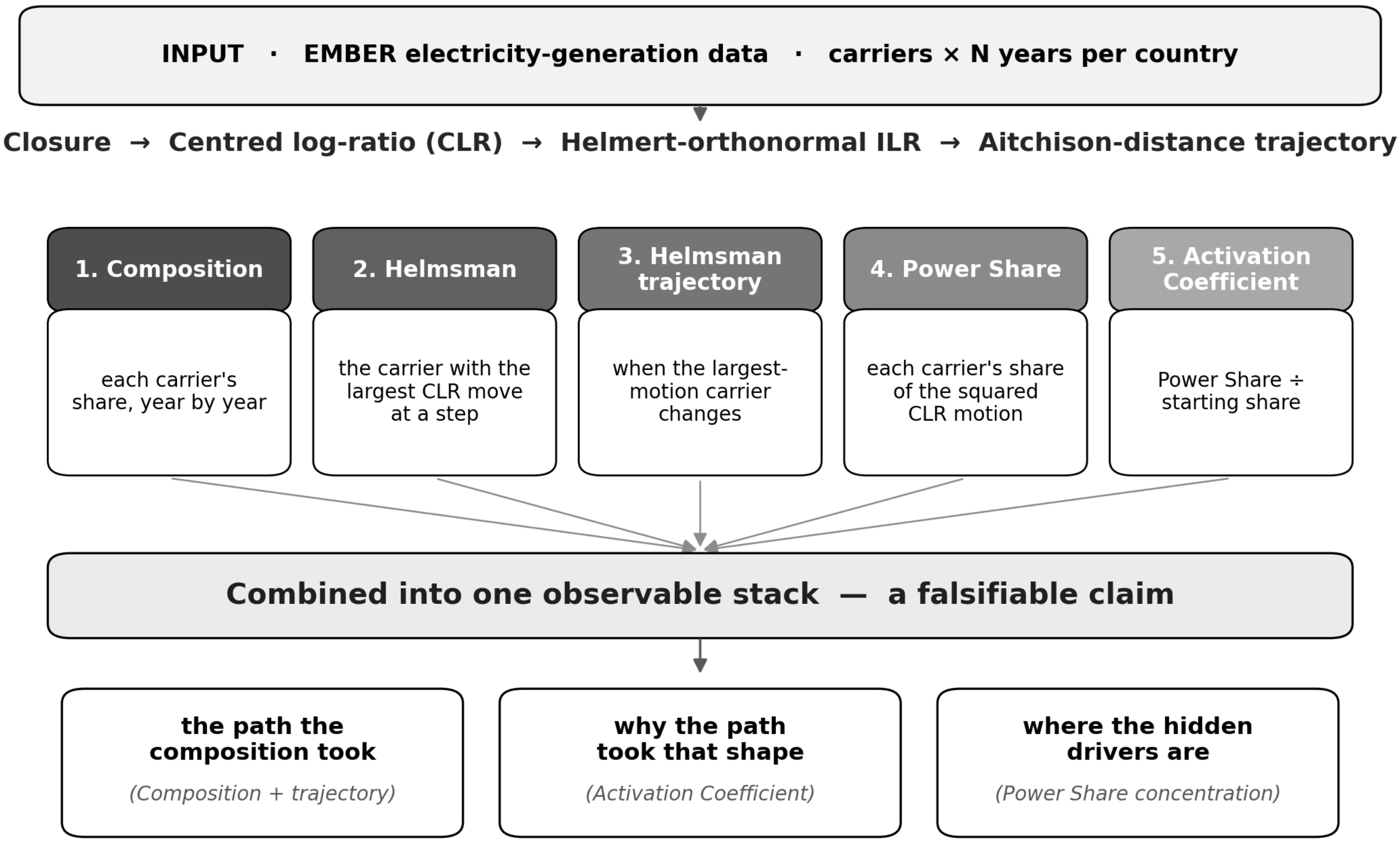
Solar acted at 333 × its size,
four years before the share view
calls solar visible.

Deceptive drift: the size view hides which carrier did the structural work — here a 0.21 % carrier does 71 % of it.

Mathematics: standard compositional data analysis. Application: monitoring frame may be new.

Reading time on the simplex

Standard CoDa geometry, read forward in time — five named readings, each defined below.



Five readings · one observable stack · reproducible from the raw CSV

The Activation Coefficient

How much structural work a carrier does, relative to how much of the mix it is.

$$\alpha_i(t) = \text{Power Share}_i(t) \div \text{starting share}_i(t)$$

$\alpha \approx 1$ carrier does work proportional to its size — ordinary

$\alpha \gg 1$ carrier acts far above its size — hidden driver

$\alpha < 1$ carrier carries less work than its size suggests — coasting

Worked example — Germany Solar 2005 → 2006

```
starting share      0.21 %     small
Power Share         71.1 %     most of the work
α                   ≈ 333 ×     hidden driver
```

Analogy — yeast is 2 % of a loaf by mass and does 100 % of the rising: the same shape.

The Energiewende's structural beginning — four years before solar appears in the share view.

Three archetypes — one instrument, three regimes

The same protocol on three transitions that look fundamentally different.

Germany

deliberate transition

continuous arc

Energiewende
2000 → 2025
solar + wind absorb
structural work
before size dominates.

Japan

external shock

loop and reorganise

Fukushima 2011
displaces nuclear,
causes 2011–2013
multi-year
reorganisation.

United Kingdom

regime change

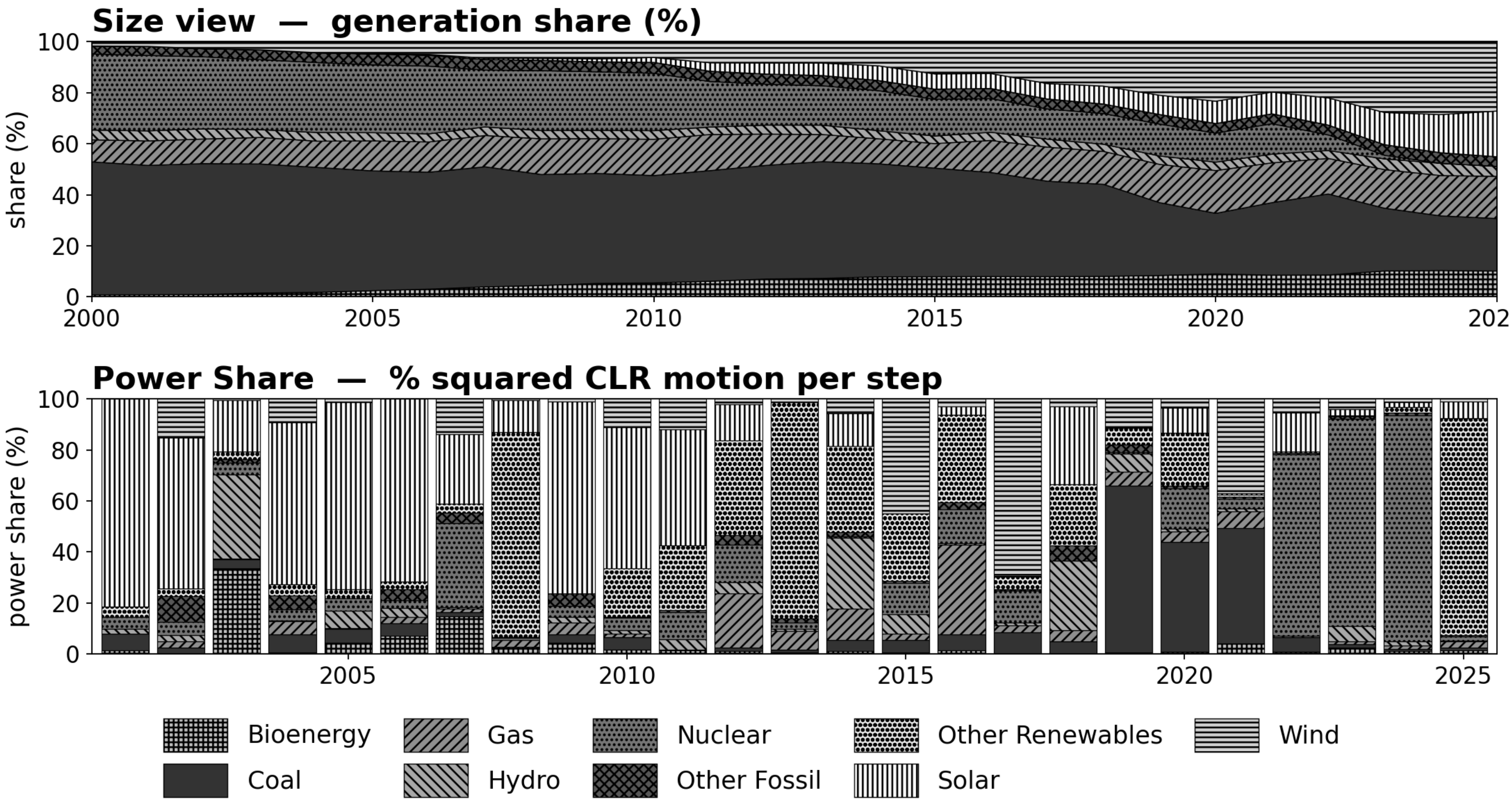
jump and return

Coal exit 2012–2020
from > 30 % to < 2 %.
Wind, solar, others
absorb displaced
structural work.

Three different transition regimes. One operational protocol reads them all.

Germany — share and structural work

The Energiewende read as a single smooth arc on the simplex.

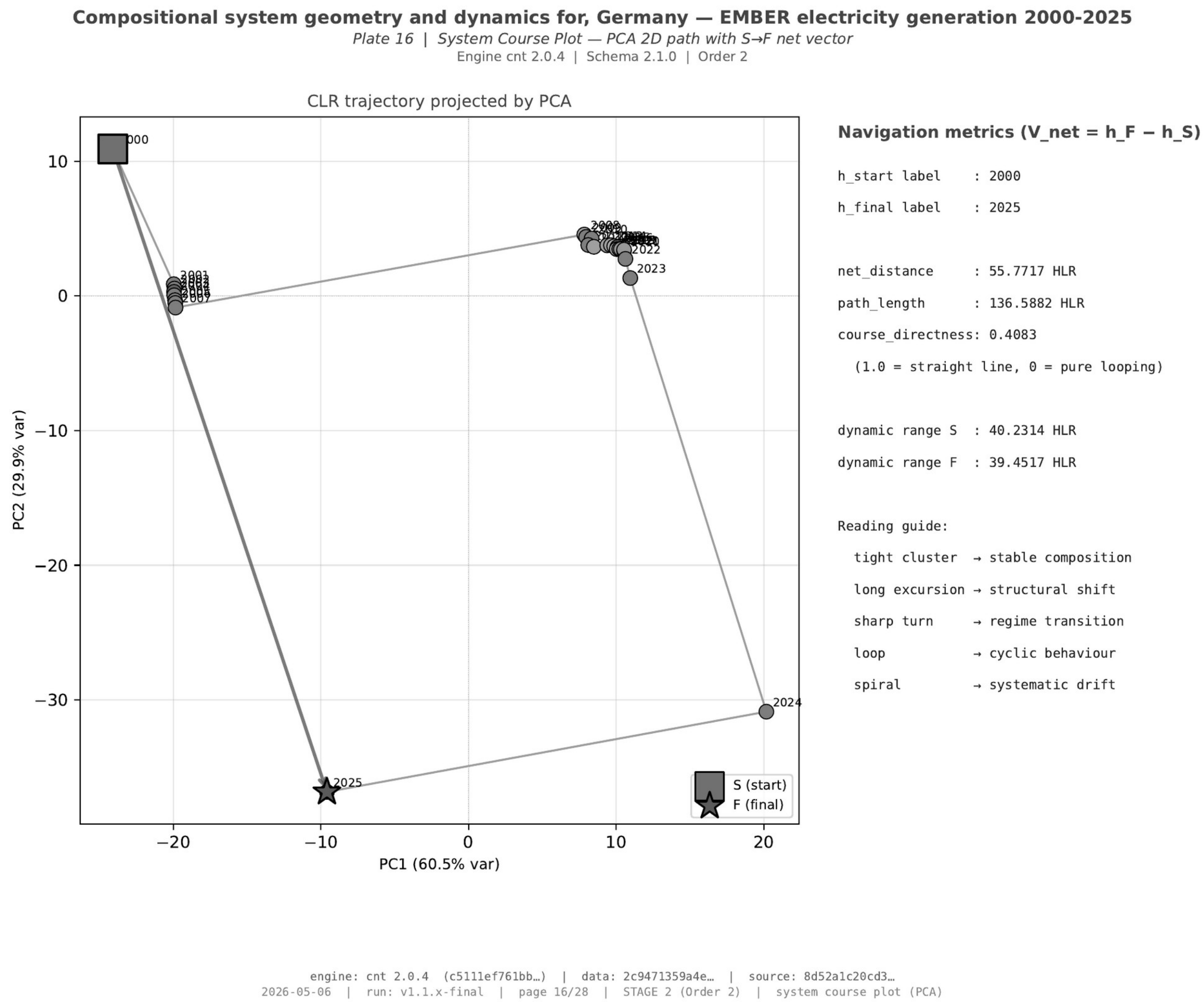


Solar 2005–2006: 0.21 % share · 71.1 % structural work · α ≈ 333 ×

Tiny share, dominant work — the deceptive-drift signature, four years before the share view shows anything.

Germany — the trajectory

The Helmsman trajectory: where the composition went, year by year.

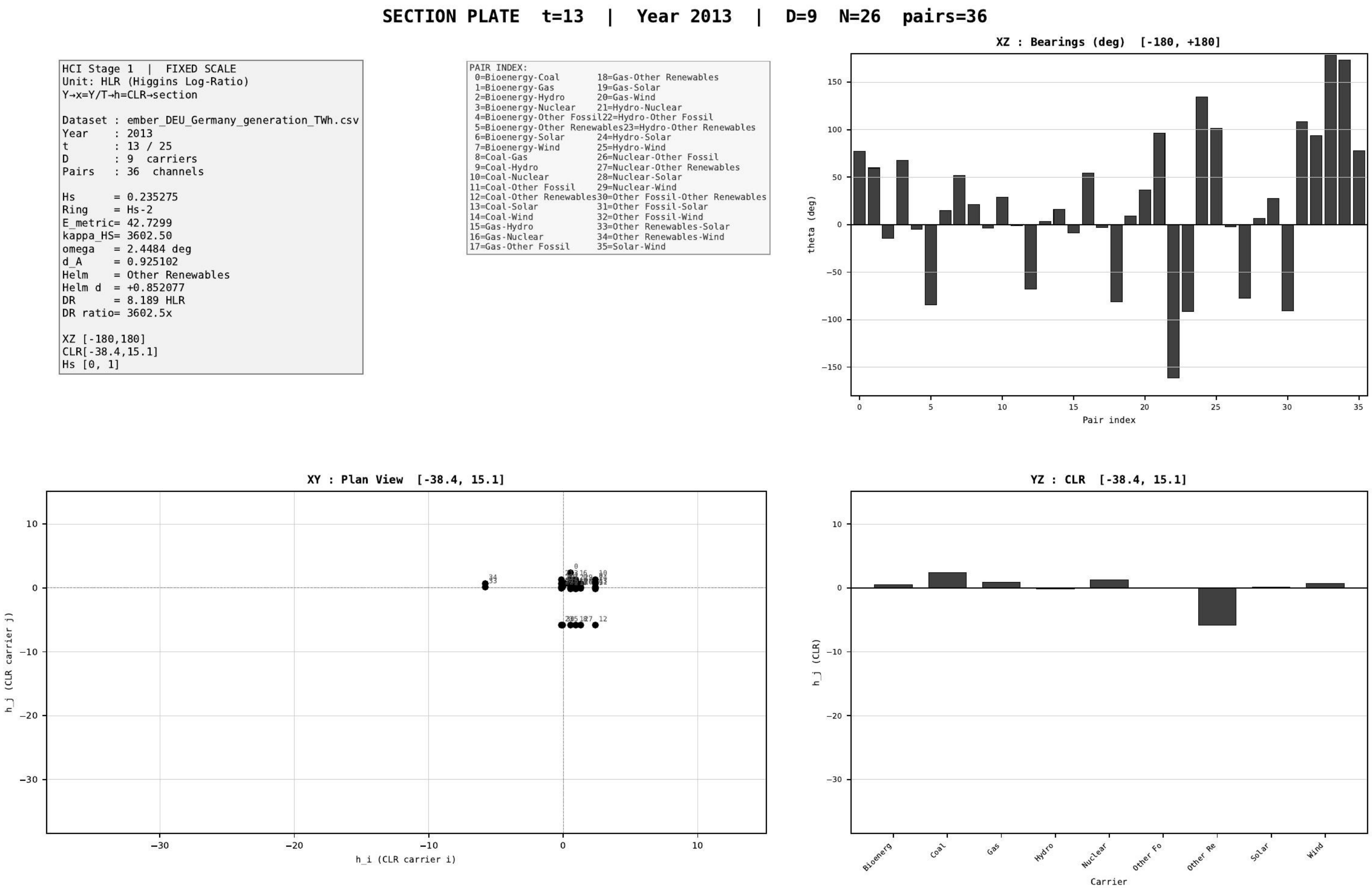


Trajectory directness 0.41 (end-to-end distance ÷ path length) — a continuous arc toward the renewable vertex.

Smooth, monotone reorientation — no loops, no flips. Deliberate transition as a sustained trajectory.

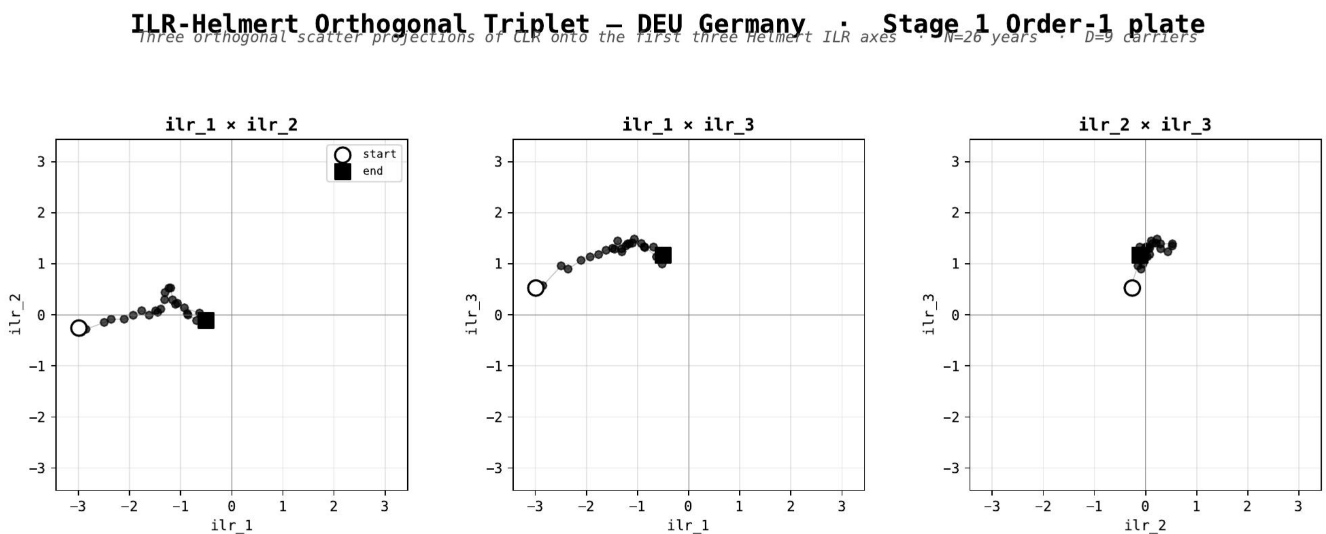
Germany — orthogonal projections (XY / XZ / YZ)

Germany alone gets the complete plate set · the Section view: CLR plan, XZ bearings, and per-carrier CLR for a representative year.



Germany — ILR-Helmert triplet

Germany's complete set · three orthonormal ILR projections of the 2000–2025 trajectory (ilr1×ilr2 · ilr1×ilr3 · ilr2×ilr3).



Helmert basis loadings (which carriers each ILR axis contrasts):

```
ilr_1:  (Bioenergy)  vs  Coal
ilr_2:  (Bioenergy + Coal)  vs  Gas
ilr_3:  (Bioenergy + Coal + Gas)  vs  Hydro
```

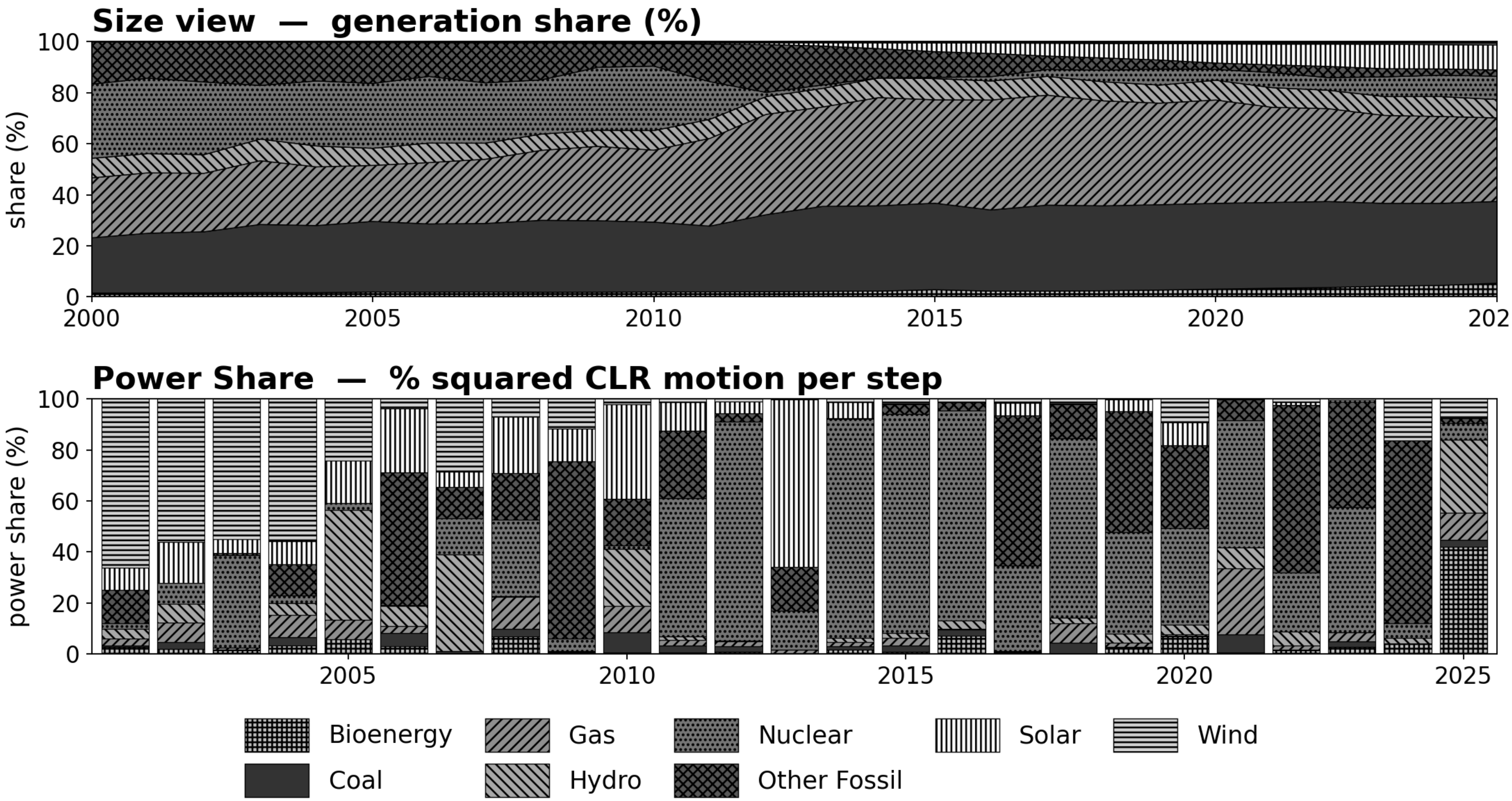
```
Trajectory:
  start year:  2000
  end year:    2025
  N readings:  26

Display:
  fixed scale ±3.44 ILR
  Order 1 (first principles)
  per Output Doctrine v1.0

Symbology:
  ○  start  ·  ▪  end
  ·  intermediate years
```

Japan — share and structural work

Fukushima 2011: an external shock displaces nuclear; a multi-year reorganisation follows.

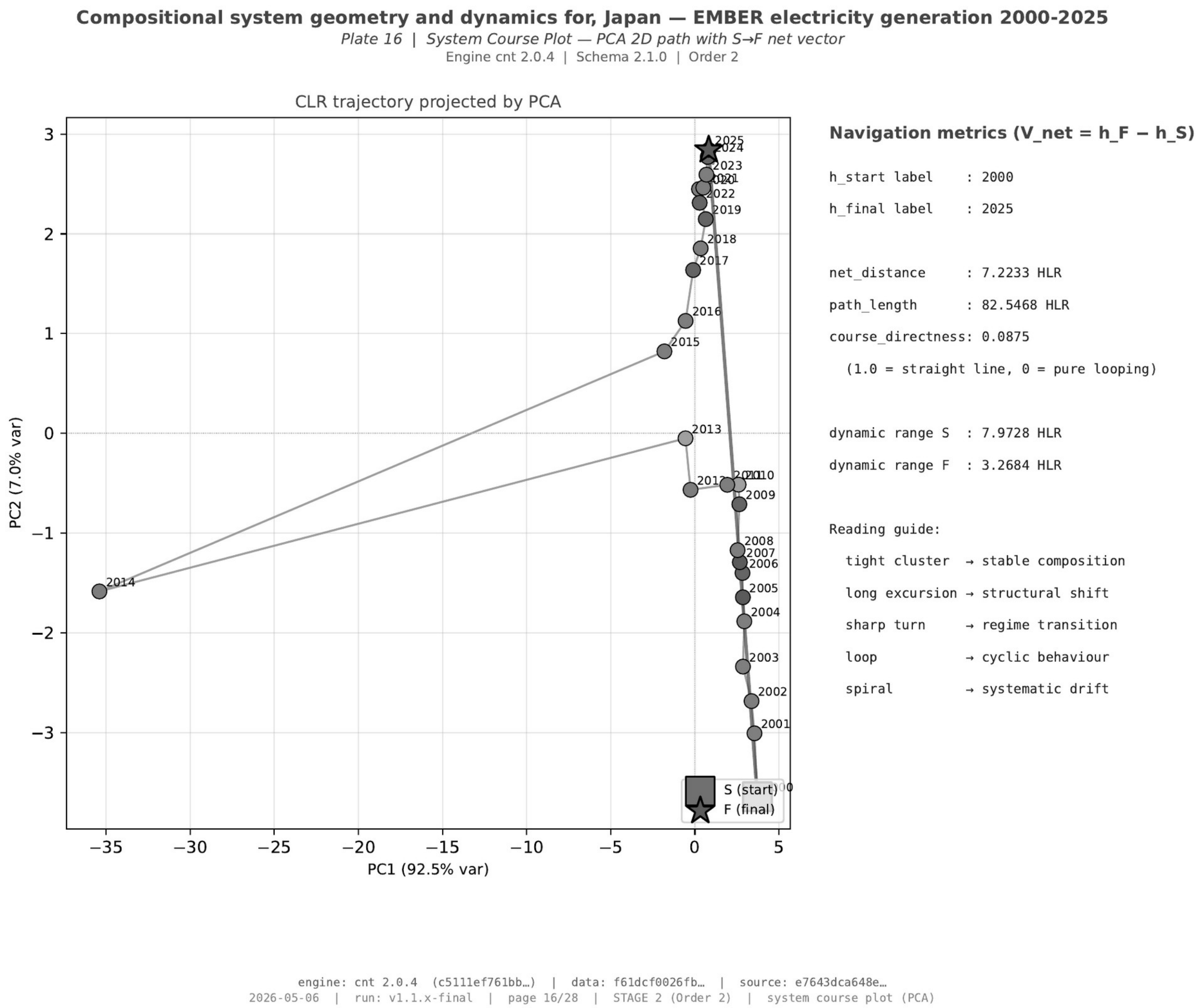


Aitchison distance (distance on the simplex) 2011 → 2012 ≈ 3 × the baseline step · 17 Helmsman flips

The instrument detects both the shock and the multi-year reorganisation that followed.

Japan — the trajectory

The Helmsman trajectory: looping reorganisation, not a single step.

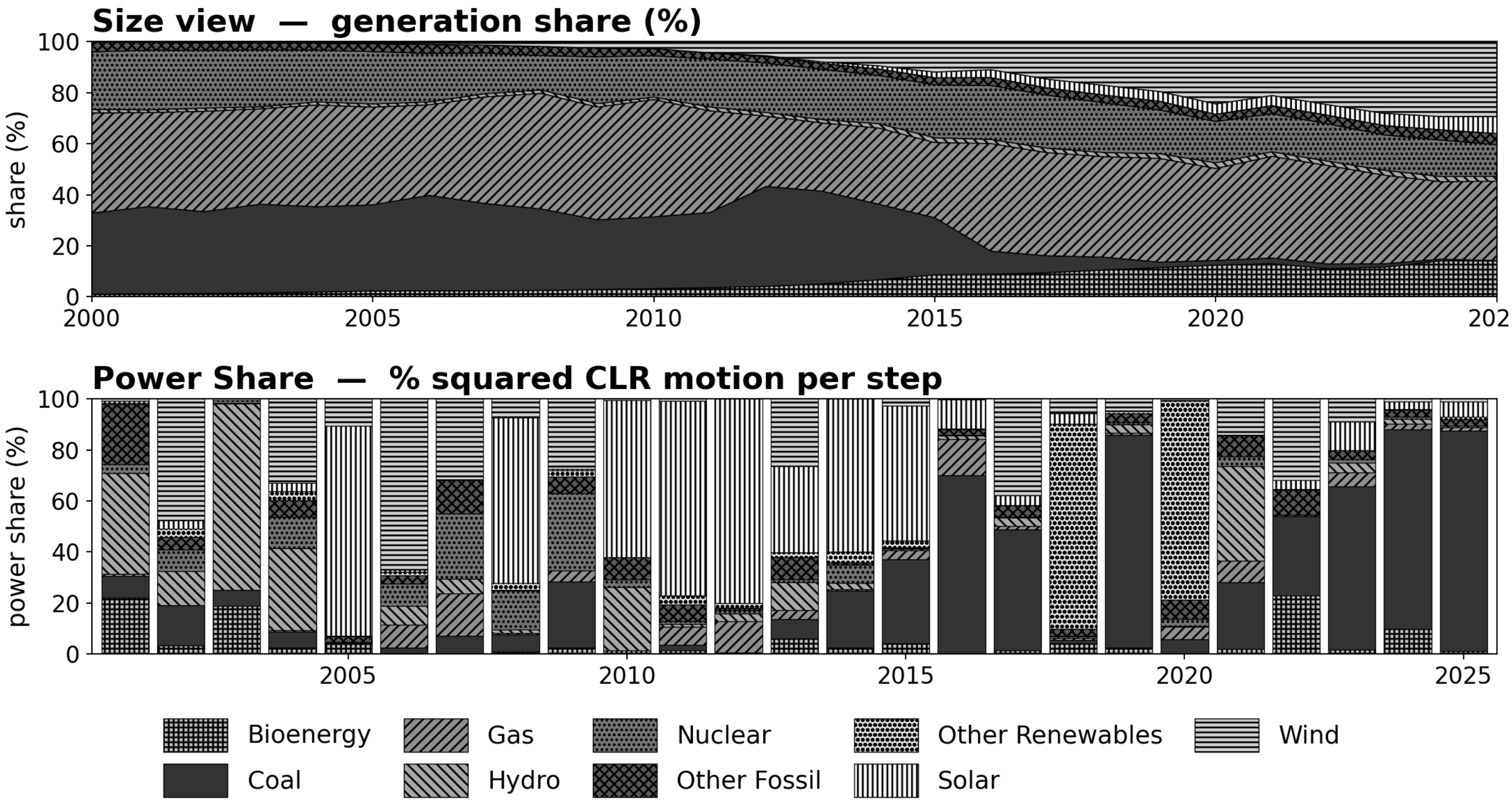


Trajectory directness 0.09 — the loop-and-reorganise pattern.

The trajectory revisits and reroutes — the system searches for a new composition, not a planned trajectory.

United Kingdom — share and structural work

Coal exit as policy-driven regime change, absorbed across several carriers.

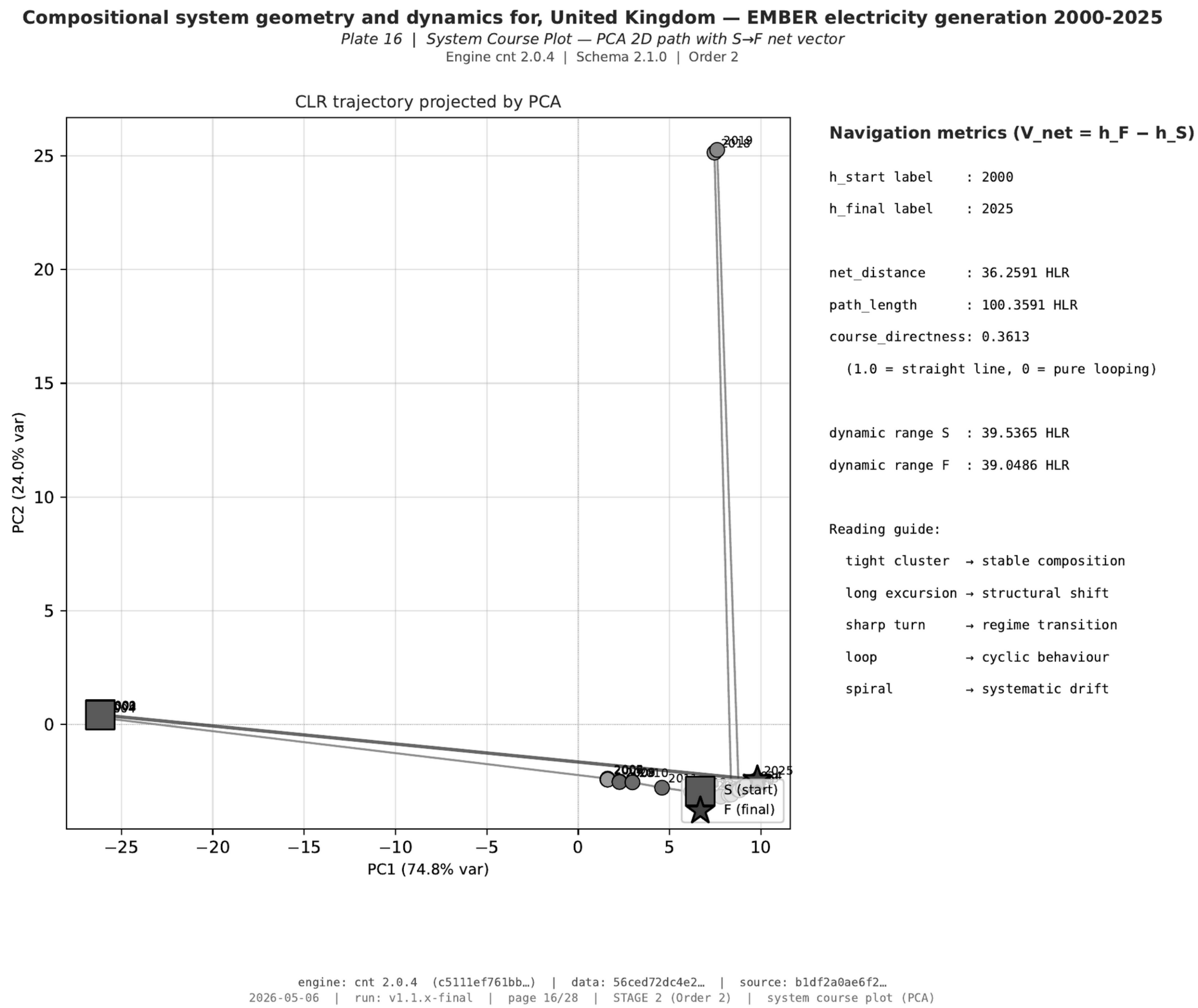


Coal: > 30 % → < 2 %. Wind, solar, and other renewables absorb the displaced work.

Displaced work redistributes across several carriers — not one substitute, but several.

United Kingdom — the trajectory

The Helmsman trajectory: a jump from the coal vertex, then a return toward stability.



Trajectory directness 0.36 — the jump-and-return pattern.

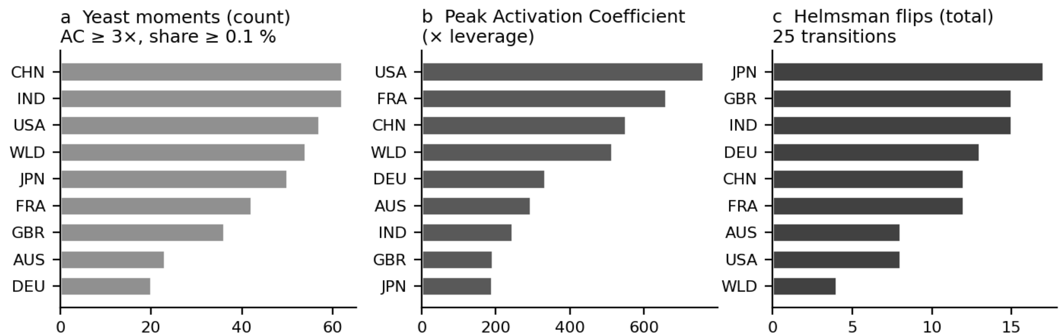
A sharp departure from the coal vertex, then re-stabilisation — regime change as one decisive displacement.

Across the corpus — deceptive drift in 5 of 9

The same protocol applied to all nine EMBER countries.

Deceptive drift: the size (share) view hides which carrier did the structural work — a small carrier carries a large share of the squared compositional motion (Activation Coefficient ≫ 1).

Detected in five of nine countries; absent in four — a detector that reads real structure must not flag every system.



Japan tops helmsman flips (17 — the shock). China & India top yeast count (62 each — small carriers absorbed). USA, France, World aggregate top peak Activation Coefficient — solar 2010–2015 at 500–760 ×.

d Top ten activation moments across the 9-country corpus

#	country	transition	carrier	Activation	Power Share	Size at start
1.	USA	2012-2013	Solar	760 ×	81.7 %	0.107 %
2.	FRA	2010-2011	Solar	659 ×	72.6 %	0.110 %
3.	FRA	2004-2005	Wind	634 ×	66.0 %	0.104 %
4.	CHN	2013-2014	Solar	549 ×	84.7 %	0.154 %
5.	WLD	2010-2011	Solar	513 ×	77.7 %	0.151 %
6.	FRA	2005-2006	Wind	495 ×	83.5 %	0.169 %
7.	USA	2013-2014	Solar	395 ×	87.9 %	0.223 %
8.	CHN	2004-2005	Bioenergy	346 ×	39.7 %	0.115 %
9.	CHN	2003-2004	Bioenergy	342 ×	45.5 %	0.133 %
10.	FRA	2024-2025	Coal	342 ×	62.1 %	0.182 %

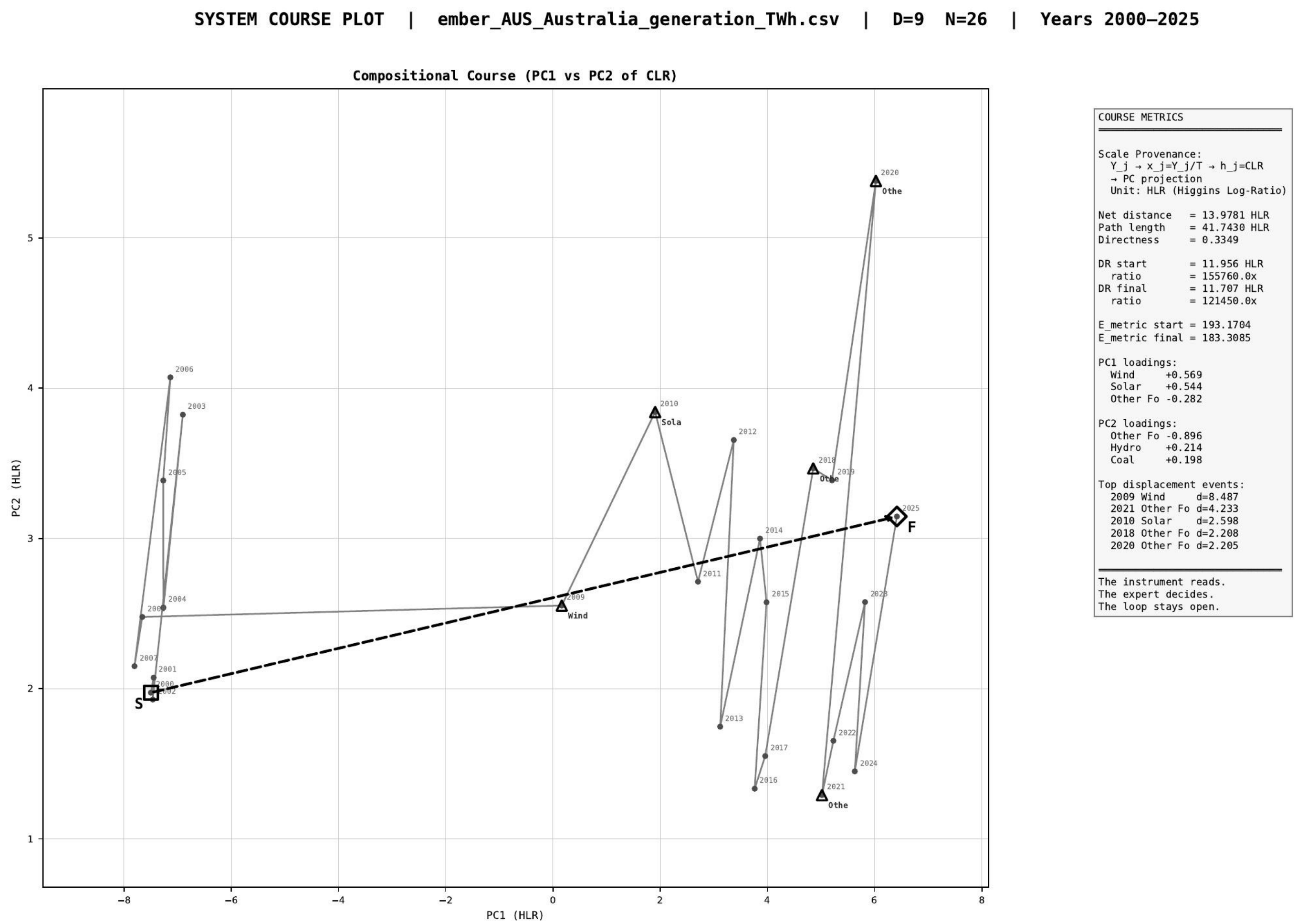
deceptive drift present: AUS · CHN · GBR · IND · JPN

deceptive drift absent: DEU (annual) · FRA · USA · WLD

Discrimination — flagging some systems and not others — is itself evidence the detector reads real structure.

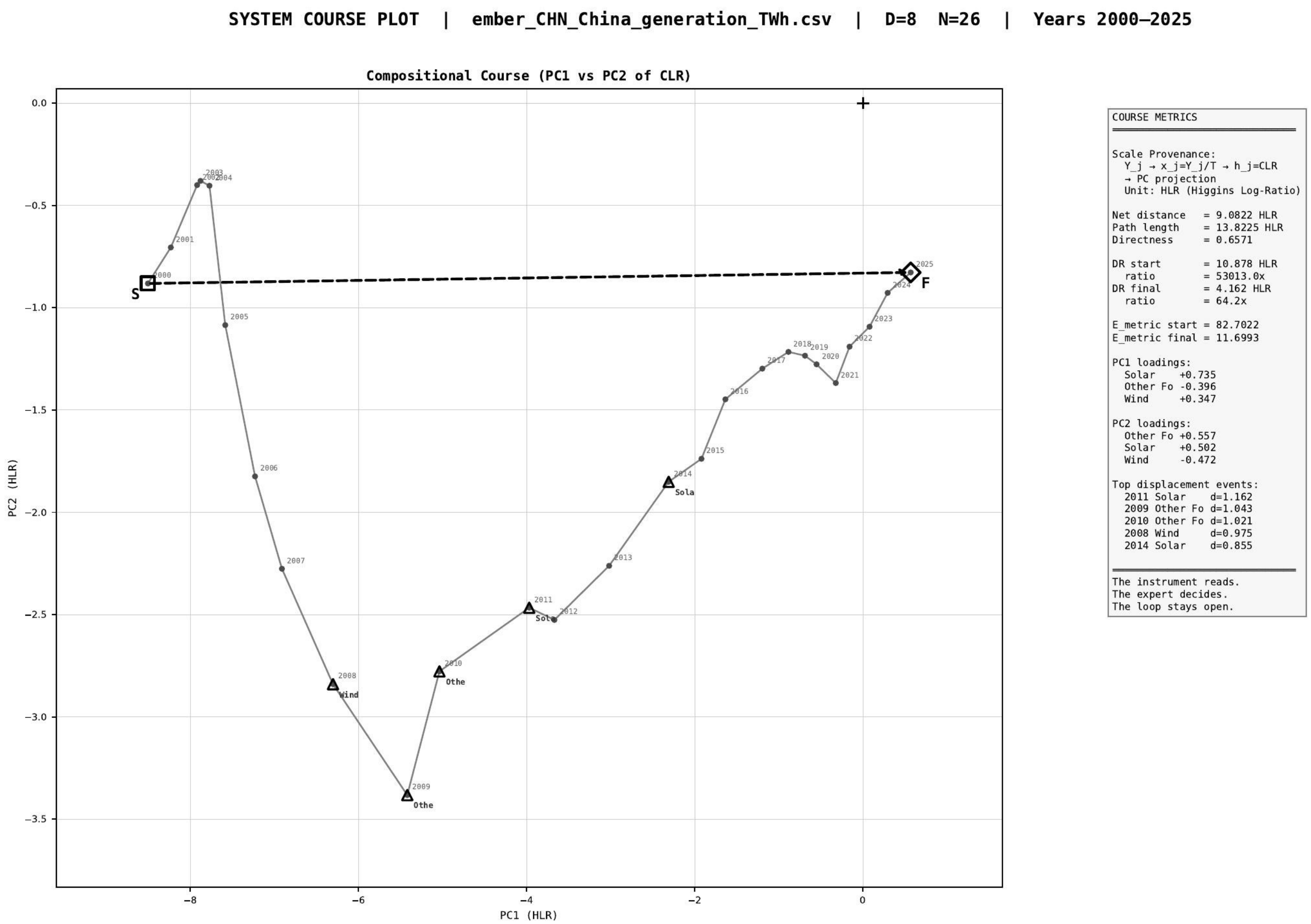
Australia — the trajectory

The rest of the world · deceptive drift present. Australia: the full 2000–2025 trajectory.



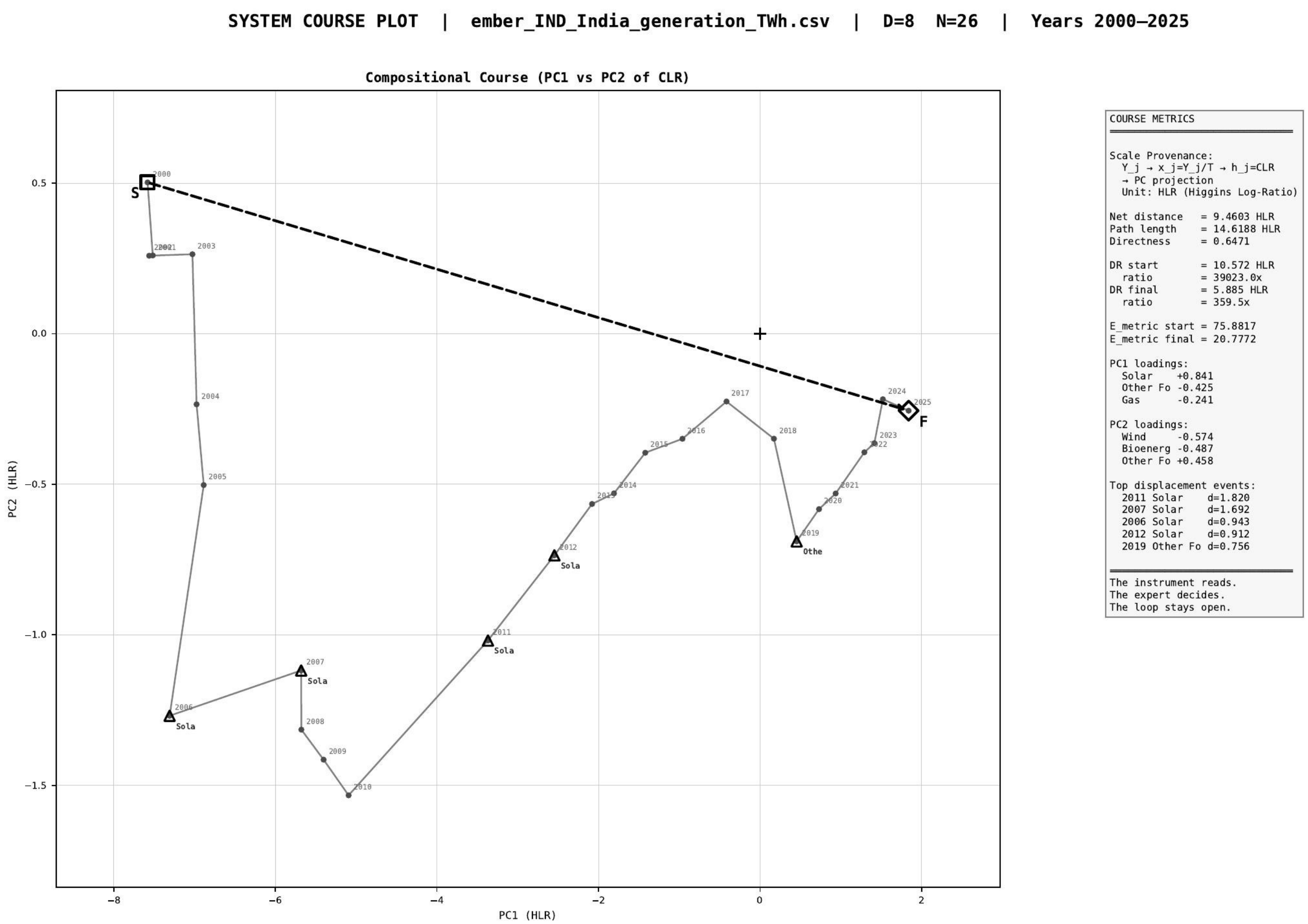
China — the trajectory

China: the full trajectory — coal-era growth, then a turn.



India — the trajectory

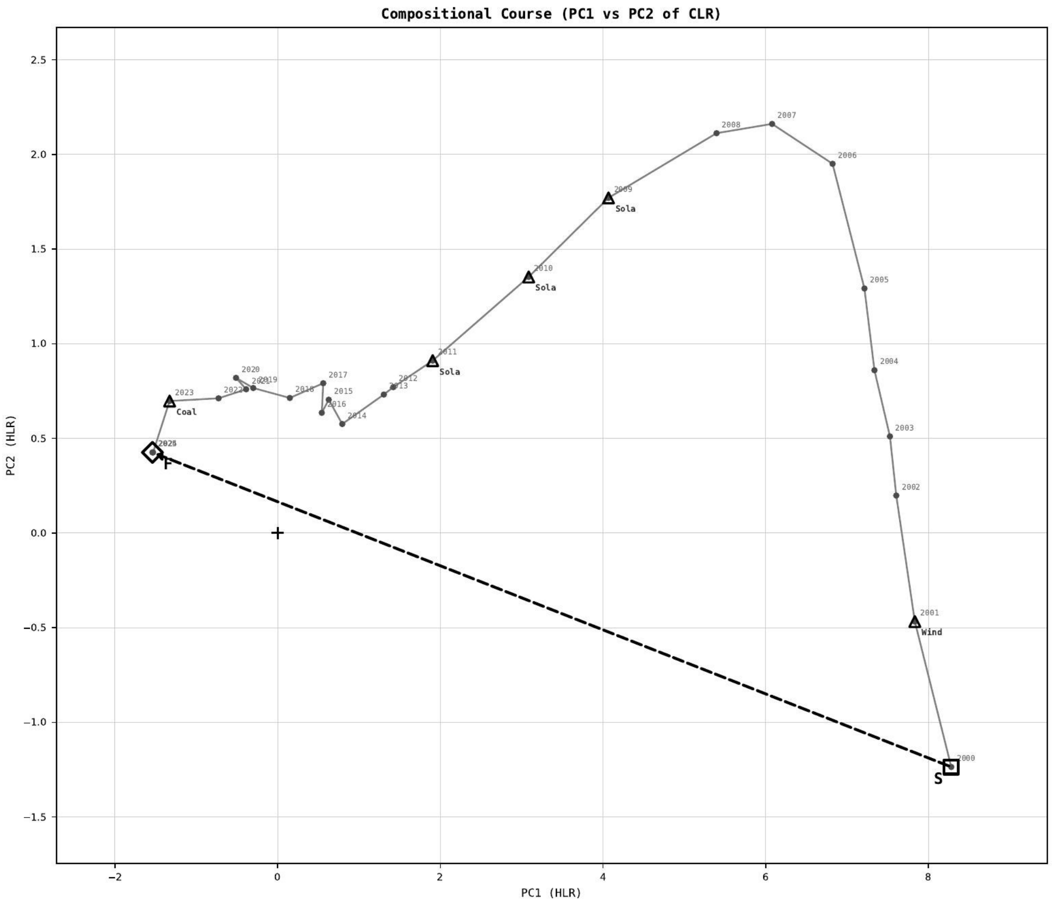
India: the full trajectory — the third case where deceptive drift is present.



France — the trajectory

Deceptive drift absent. France: the full trajectory — nuclear-stable.

SYSTEM COURSE PLOT | ember_FRA_France_generation_TWh.csv | D=9 N=26 | Years 2000–2025

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COURSE METRICS
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Scale Provenance:
  Y_j → x_j=Y_j/T → h_j=CLR
  → PC projection
  Unit: HLR (Higgins Log-Ratio)

Net distance   = 10.0035 HLR
Path length    = 17.9314 HLR
Directness     = 0.5579

DR start       = 10.634 HLR
  ratio        = 41516.0x
DR final       = 6.534 HLR
  ratio        = 687.8x

E_metric start = 94.4242
E_metric final = 29.9972

PC1 loadings:
  Solar    -0.768
  Coal     +0.387
  Wind     -0.385

PC2 loadings:
  Wind     +0.841
  Solar    -0.448
  Hydro    -0.160

Top displacement events:
  2009 Solar    d=1.437
  2011 Solar    d=1.337
  2010 Solar    d=1.129
  2023 Coal     d=1.095
  2001 Wind     d=0.998

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The instrument reads.
The expert decides.
The loop stays open.
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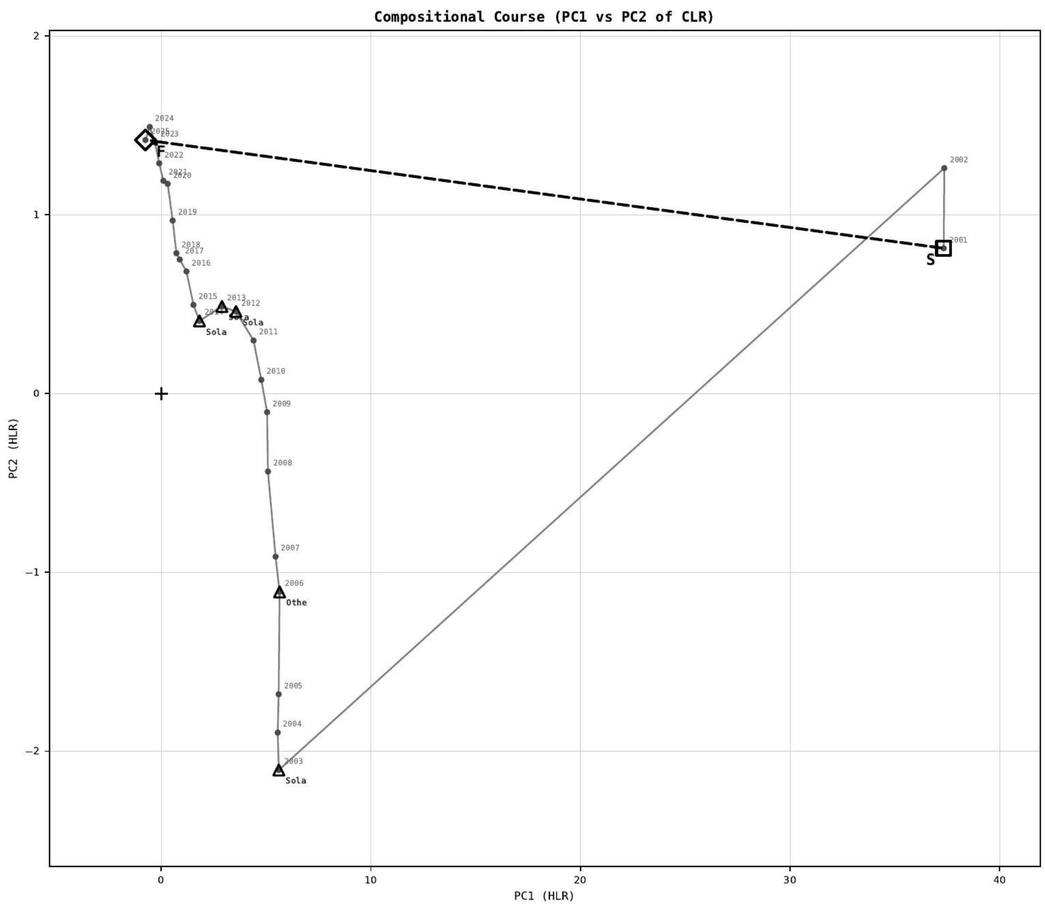
United States — the trajectory

United States: the full trajectory — a large grid; no deceptive drift at annual grain.

SYSTEM COURSE PLOT | ember_USA_United_States_generation_TWh.csv | D=9 N=25 | Years 2001–2025

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COURSE METRICS
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Scale Provenance:
  Y_j → x_j=Y_j/T → h_j=CLR
  → PC projection
  Unit: HLR (Higgins Log-Ratio)

Net distance   = 38.0885 HLR
Path length    = 41.8379 HLR
Directness     = 0.9104

DR start       = 42.090 HLR
  ratio        = 1903969992697630976.0x
DR final       = 4.633 HLR
  ratio        = 102.8x

E_metric start = 1411.9935
E_metric final = 20.9704

PC1 loadings:
  Solar    -0.938
  Other Fo +0.159
  Coal     +0.146

PC2 loadings:
  Wind     +0.831
  Other Fo -0.385
  Coal     -0.324

Top displacement events:
  2003 Solar    d=31.925
  2014 Solar    d=1.080
  2012 Solar    d=0.882
  2006 Other Fo d=0.708
  2013 Solar    d=0.691

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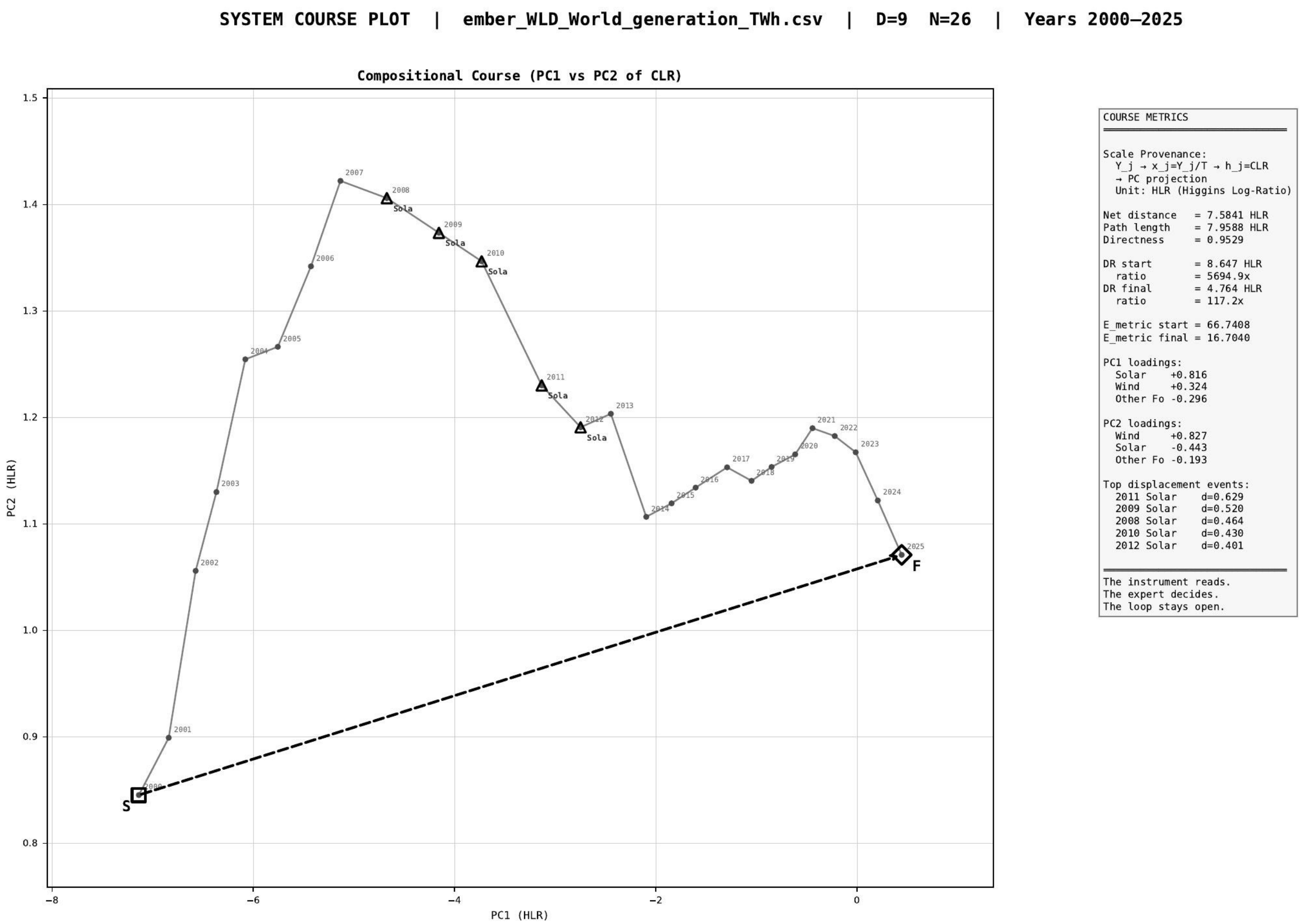
The instrument reads.
The expert decides.
The loop stays open.
```

World — the trajectory

World: the aggregate trajectory — large-N smoothing hides the deceptive drift in its constituents.



What the stack answers

Recap of the five readings — then the live instrument takes over.

WHAT	carriers are big.	*size view*
WHO	is in largest motion.	*Helmsman*
WHEN	the largest-motion carrier changes.	*Helmsman trajectory*
HOW MUCH	structural work each carrier did.	*Power Share*
WHY	a small carrier mattered.	*Activation Coefficient*

The stack does not replace interpretation. It gives interpretation a reproducible object.

The live instrument takes it from here → codawork2026_projector.html

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